

Who Does the Recommender Serve? A Fairness Audit and Intersectional Mitigation of Demographic Bias in Distributed Rating Classification on MovieLens 1M

Shahriar Fahim, Rutvik Katkoriya, Mostafizur Rahman, Aliia Rustamova

Department of Computer Science

San Francisco Bay University

Fremont, CA, USA

Abstract—Rating-prediction models are routinely evaluated on aggregate accuracy, which hides whether a model serves different demographic groups equally. We audit a distributed logistic-regression classifier that predicts whether a user rates a movie highly (rating ≥ 4) on the MovieLens 1M dataset, implemented as an Apache Spark ML pipeline. We first identify and correct a target-leakage flaw in a common feature recipe—user- and movie-average ratings computed over the full dataset—which modestly inflated accuracy from 0.723 to a corrected 0.719; correcting it is nonetheless a prerequisite for an honest audit. On the corrected model, aggregate accuracy of 0.72 conceals sharply different behavior by protected attribute: gender disparity is mild (demographic parity difference, DPD, of 0.036; disparate impact, DI, of 0.95), but age disparity is substantial (DPD = 0.146, DI = 0.80, at the common four-fifths threshold), with the model under-selecting items for younger users and over-selecting for older users. We then evaluate mitigations. Removing the protected attributes from the feature set (“fairness through unawareness”) did not help and significantly worsened the age gap (DPD 0.145 $\rightarrow$ 0.213, $p < 10^{-10}$), because correlated proxies remain. Single-attribute per-group threshold calibration eliminates the *targeted* gap (gender DPD $\rightarrow$ 0.001) at negligible accuracy cost, but does not transfer to the other attribute. We therefore introduce *intersectional* threshold calibration, which sets a decision threshold per gender $\times$ age cell and reduces *both* the gender and age gaps together (DPD 0.038 $\rightarrow$ 0.001 and 0.145 $\rightarrow$ 0.003) at a 0.21-point accuracy cost. All headline results are validated across eight random splits with paired significance tests. Our findings argue that fairness must be measured and mitigated per attribute—jointly, not one at a time—that dropping protected features is insufficient, and that leakage-aware evaluation is a precondition for trustworthy fairness claims.

Index Terms—algorithmic fairness, recommender systems, Apache Spark, logistic regression, demographic bias, MovieLens, disparate impact, intersectional fairness, threshold calibration, significance testing

I. INTRODUCTION

Recommender and rating-prediction systems increasingly mediate what content users see, yet their evaluation is dominated by aggregate accuracy that treats all users as interchangeable. A model can report strong overall accuracy while systematically under-serving a demographic subgroup. As machine

learning moves into consequential settings, this gap between aggregate performance and per-group performance has become central to trustworthy AI [1], [2].

We study this on a concrete, reproducible artifact: a logistic-regression classifier predicting the binary target “high rating” (rating ≥ 4) for a (user, movie) pair on MovieLens 1M [3], built as an Apache Spark ML pipeline [4], [5]. The setting is deliberately mainstream—logistic regression on MovieLens is a canonical baseline—which makes it a useful lens on a broader question: *when a standard, well-performing pipeline is deployed at scale, who does it actually serve?*

Three findings organize the paper. First, the pipeline as originally constructed suffered from *target leakage*: user- and movie-average rating features were computed over the entire dataset, including the ratings being predicted. Correcting this changes accuracy only modestly (0.723 $\rightarrow$ 0.719), but it is a precondition for an honest audit. Second, the corrected model’s aggregate accuracy of 0.72 conceals a strong asymmetry between protected attributes: gender disparity is mild, whereas age disparity is large and sits at the four-fifths disparate-impact threshold. Third, the intuitive mitigation of removing protected attributes fails and even worsens the age gap, while a cheap *single-attribute* per-group threshold calibration removes the targeted gap almost entirely but does not transfer across attributes—motivating an *intersectional* calibration that closes both gaps at once. Every headline claim is validated across eight random splits with paired significance tests, so the disparities and the mitigation effects are not single-seed artifacts.

Contributions.

- We identify and correct a target-leakage flaw in a common MovieLens feature recipe and quantify its (modest but statistically consistent) inflationary effect (§III-C, §V).
- We conduct a group-fairness audit of a scalable Spark logistic-regression classifier across gender and age using DPD, equal-opportunity difference (EOD), and disparate

impact, showing that a single aggregate score hides an attribute-dependent disparity (§V).

- We show that “fairness through unawareness” *significantly worsens* the age gap, and that single-attribute threshold calibration does not transfer across protected attributes (§VI).
- We introduce *intersectional threshold calibration*—a lightweight post-processing method that equalizes selection rate across gender×age cells—and show it reduces both gender and age disparity together at a small accuracy cost (§VI).
- We validate all headline results across eight random splits with paired *t*- and Wilcoxon tests, and release a reproducible Spark notebook (§IV, §VI).

II. RELATED WORK

Fairness definitions. Dwork et al. introduced fairness through awareness [6]; Hardt et al. proposed equalized odds and equal opportunity, framing fairness as error-rate parity across groups [7]; Feldman et al. formalized disparate impact and its removal [8]. Verma and Rubin catalog these often-conflicting definitions [9]. We report demographic parity, equal opportunity, and disparate impact together rather than committing to one.

Bias in recommendation. Ekstrand et al. show recommender effectiveness and evaluation can differ across demographic groups [10], and surveys document how bias enters at data, model, and evaluation stages [1]. Most such work targets ranking or top-*k* recommendation; we audit the upstream rating-classification stage, where disparities originate but are rarely measured.

Mitigation. Mitigations are pre-, in-, or post-processing. Kamiran and Calders study pre-processing via reweighing [11]; post-processing adjusts decision thresholds per group [7]. We evaluate one pre-processing approach (removing sensitive attributes) and a family of post-processing approaches (single-attribute and *intersectional* per-group thresholds), all interpretable and compatible with a distributed pipeline. Our feature-removal result is a concrete instance of the well-known limitation of “fairness through unawareness” [6]; our intersectional calibration extends standard single-attribute thresholding [7] to the joint attribute cell so that multiple demographic gaps are closed together rather than one at a time.

Gap we address. Prior fairness studies rarely (i) operate on a distributed big-data engine, (ii) tie the audit to a reproducible interpretable model, and (iii) control for evaluation artifacts such as target leakage. We address all three.

III. METHODOLOGY

A. Dataset

MovieLens 1M [3] contains 1,000,209 ratings from 6,040 users on ~3,900 movies, with per-user gender, age band, and occupation. We load the “::”-delimited files with explicit schemas, drop duplicate (user, movie) ratings, enforce referential integrity, and restrict ratings to [1, 5].

B. Task and features

The target is $y = \mathbb{I}[\text{rating} \geq 4]$. Features are (i) *demographic* (age, occupation, gender), (ii) *static movie* attributes (release year, movie age relative to 2000, number of genres), and (iii) *derived interaction* statistics (user-average rating, movie-average rating, user rating count, movie popularity). Gender and age serve both as baseline features and as the *protected attributes* for the audit.

C. Leakage-aware feature construction

The derived interaction statistics are label-dependent: a user’s average rating over all their ratings includes the rating being predicted. Computing them over the full dataset before splitting leaks the test label into test features. We correct this by performing the 80/20 split *first* (seed 42) and computing all derived statistics on the training split only, joining them onto test rows and backing off unseen users/movies to the global training mean (3.582). We also build a leaky reference model (statistics over all data) to measure the effect.

D. Model

The classifier is a Spark ML Pipeline of `VectorAssembler`→`StandardScaler`→`LogisticRegression` (max 100 iterations). Logistic regression is chosen for interpretability: coefficients expose which features drive predictions.

E. Fairness metrics

For a protected group g , let the selection rate be $s_g = P(\hat{y} = 1 \mid g)$ and the true-positive rate $t_g = P(\hat{y} = 1 \mid y = 1, g)$. We report

$$\text{DPD} = \max_g s_g - \min_g s_g, \quad \text{DI} = \min_g s_g / \max_g s_g, \quad (1)$$

$$\text{EOD} = \max_g t_g - \min_g t_g. \quad (2)$$

DPD and DI capture demographic parity; EOD captures equal opportunity. A DI below 0.8 is a common disparate-impact threshold [8].

F. Mitigations

We evaluate one pre-processing and a family of post-processing mitigations, all interpretable and compatible with a distributed pipeline.

Sensitive-feature removal retrain with gender and age excluded from the feature set (“fairness through unawareness”).

Threshold calibration. We keep the baseline model’s scores and, for each subgroup g defined by one or more protected attributes, choose the decision threshold t_g equal to the empirical $(1 - \tau)$ quantile of the positive-class score within g , where τ is the overall target selection rate. By construction $P(\hat{y} = 1 \mid g) \approx \tau$ for every g , driving the selection-rate gap toward zero. When the subgroup is defined by a *single* attribute (gender or age), this is standard per-group post-processing [7]. Our *intersectional threshold calibration* instead defines subgroups by the *joint* gender×age cell (14 cells here), so a single pass equalizes selection rate across all cells

TABLE I
EFFECT OF LEAKAGE CORRECTION (TEST SET, $n = 199,705$).

Model	Accuracy	Precision	Recall	F1
Leaky (full-data averages)	0.723	0.731	0.821	0.773
Corrected (train-only)	0.719	0.727	0.817	0.770

and hence reduces *both* marginal demographic-parity gaps simultaneously—at the cost of estimating more thresholds on smaller strata. We compare single-attribute (gender-only, age-only) against intersectional calibration to test whether joint conditioning is necessary.

IV. EXPERIMENTAL SETUP

Experiments run on a single machine in Spark local mode (`local[*]`), Spark 3.5.3, Java 17, Python 3.11. For the detailed per-group tables we report the seed-42 split (training set 800,504 rows, test set 199,705). The primary accuracy metric is overall accuracy; the primary fairness metric is gender DPD, complemented by age DPD, EOD, and DI. We audit gender (two groups) and age (seven MovieLens bands). Reported F1 is for the positive class.

Cross-seed validation. To ensure results are not artifacts of one split, we repeat the entire pipeline—leakage-corrected retraining, audit, and all mitigations—over eight independent random 80/20 splits (seeds {42, 1, 7, 13, 21, 99, 123, 2024}) and report mean $\pm$ standard deviation. For each directional claim we run a *paired*, seed-matched Student’s *t*-test (one-sided) and the non-parametric Wilcoxon signed-rank test as a robustness check; the “no-transfer” claim is tested two-sided, where non-significance is the expected outcome.

V. FAIRNESS AUDIT

A. Leakage correction

Table I reports the leaky and corrected models. Correcting the leakage moved accuracy from 0.723 to 0.719 and positive-class F1 from 0.773 to 0.770. The effect is modest—each average aggregates ~ 165 ratings per user, so removing one rating barely shifts it—but the correction is methodologically necessary and all subsequent results use the corrected model.

B. Group disparities

Table II shows per-gender behavior. The gender gap is mild: DPD = 0.036, DI = 0.95 (well above the 0.8 threshold), and EOD = 0.007—by these measures the model is close to parity across gender. The picture differs sharply for age. As Fig. 1 shows, the baseline selection rate rises almost monotonically with user age, from 0.59 for the 18–24 band to 0.73 for the 56+ band, giving an age DPD of 0.146, DI of 0.80 (at the four-fifths threshold), and EOD of 0.090. Aggregate accuracy of 0.72 therefore masks a substantial age disparity while gender looks benign.

TABLE II
GENDER FAIRNESS OF THE CORRECTED BASELINE.

Group	n	Sel. rate	TPR	Accuracy
Female	49,177	0.673	0.823	0.710
Male	150,528	0.637	0.816	0.722
Gender: DPD = 0.036, EOD = 0.007, DI = 0.95				
Age: DPD = 0.146, EOD = 0.090, DI = 0.80				

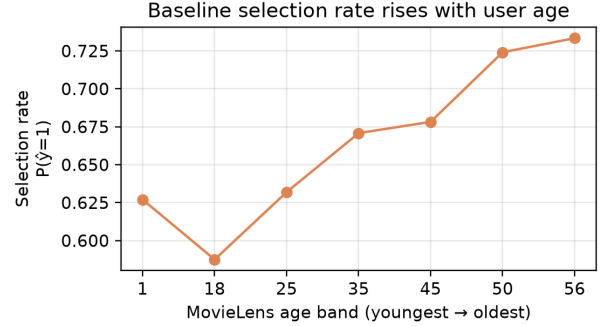


Fig. 1. Baseline selection rate $P(\hat{y} = 1)$ rises with user age band, revealing a disparity invisible in the aggregate accuracy of 0.72.

VI. MITIGATION AND TRADEOFF

Table III compares the corrected baseline with four mitigations on *both* protected attributes (seed 42), Fig. 2 visualizes the same comparison, and Table IV reports the cross-seed significance tests that back the claims below.

Feature removal significantly backfires. Dropping gender and age from the feature set did not improve gender parity (DPD 0.036 $\rightarrow$ 0.042) and *worsened* the age gap (DPD 0.146 $\rightarrow$ 0.215, DI 0.80 $\rightarrow$ 0.72). Across eight seeds the age gap rose from 0.145 ± 0.005 to 0.213 ± 0.003 , a paired increase of 0.068 that is highly significant (*t*-test $p = 1.8 \times 10^{-11}$; Wilcoxon $p = 0.004$; H3 in Table IV). Removing protected attributes leaves correlated proxies—movie age, genre mix, popularity—through which the model reconstructs, and even sharpens, the disparity: a concrete instance of the limits of fairness through unawareness [6].

Single-attribute thresholds work but do not transfer. Calibrating a per-gender threshold reduced gender DPD from 0.036 to 0.003 (DI = 0.995) at an accuracy cost of under 0.001, but left the age gap untouched (0.146 $\rightarrow$ 0.145); across seeds the gender-threshold age gap is statistically indistinguishable from baseline (H4: $p = 0.94$, i.e. *no* transfer). Symmetrically, an age-only threshold closes the age gap (0.146 $\rightarrow$ 0.003) while leaving gender unchanged (0.037). Post-processing is cheap and effective, but only for the attribute it conditions on.

Intersectional calibration closes both gaps at once. Conditioning the threshold on the joint gender $\times$ age cell (14 thresholds) reduced gender DPD to 0.001 *and* age DPD to 0.003 in a single pass. Across eight seeds it drove gender DPD from 0.038 ± 0.002 to 0.001 ± 0.000 (H1: $p = 3.4 \times 10^{-10}$)

TABLE III
FAIRNESS–ACCURACY COMPARISON ACROSS MITIGATIONS (SEED 42).
BEST (LOWEST) DPD PER ATTRIBUTE IN **BOLD**.

Configuration	Acc.	G-DPD	G-DI	A-DPD	A-DI
Corrected baseline	0.719	0.036	0.95	0.146	0.80
Feature removal	0.719	0.042	0.94	0.215	0.72
Gender threshold	0.718	0.003	0.99	0.145	0.80
Age threshold	0.718	0.037	0.95	0.003	0.99
Intersectional threshold	0.717	0.001	1.00	0.003	0.99

TABLE IV
CROSS-SEED PAIRED SIGNIFICANCE TESTS ($n = 8$ SEEDS). MEAN DIFF. IS
THE SEED-MATCHED DIFFERENCE; p_t IS THE PAIRED t -TEST, p_W THE
WILCOXON SIGNED-RANK. H1–H3 ARE ONE-SIDED; H4 IS TWO-SIDED
(NON-SIGNIFICANCE EXPECTED).

Hypothesis (on DPD)	Mean diff.	p_t	p_W
H1: baseline – intersect. (gender)	0.038	3.4×10^{-10}	0.004
H2: baseline – intersect. (age)	0.143	1.7×10^{-12}	0.004
H3: feat.-removal – baseline (age)	0.068	1.8×10^{-11}	0.004
H4: gender-thr. – baseline (age)	0.000	0.94	0.95

and age DPD from 0.145 ± 0.005 to 0.003 ± 0.001 (H2: $p = 1.7 \times 10^{-12}$), at an accuracy cost of only 0.21 points ($0.7178 \pm 0.0012 \rightarrow 0.7157 \pm 0.0011$). Fig. 3 shows the age-gap distributions across seeds: unawareness shifts the whole distribution up, single-attribute gender thresholding leaves it at baseline, and intersectional calibration collapses it to near zero with tight variance. Fig. 4 plots the gender fairness–accuracy tradeoff with per-seed error bars: both threshold methods move left (fairer) with only a small vertical (accuracy) drop, and intersectional calibration achieves this jointly for age as well.

VII. DISCUSSION, LIMITATIONS, AND THREATS TO VALIDITY

Interpretation. The audit yields four transferable lessons: (i) a single aggregate metric can be simultaneously acceptable and misleading—here gender looks fair while age does not; (ii) removing protected attributes does not remove bias and significantly worsens it via proxies; (iii) cheap post-processing closes a targeted gap but does not generalize across attributes; and (iv) conditioning the same post-processing on the *joint* attribute cell closes multiple gaps at once for a small accuracy cost.

Internal validity. Every headline result is validated across eight random splits with paired t - and Wilcoxon tests (§VI); the standard deviations are small (≤ 0.005 DPD) and all directional effects are significant at $p < 10^{-9}$, so they are not seed artifacts. Two limits remain. Intersectional calibration estimates thresholds on smaller strata (here 14 cells, the smallest with $\sim 5,400$ test rows), so it may not scale to many fine-grained attributes without regularized or smoothed thresholds. Per-group thresholding also assumes group membership is available at inference, which raises governance questions of its own.

External validity. MovieLens users are not a representative population, and gender is a self-reported binary; the audit

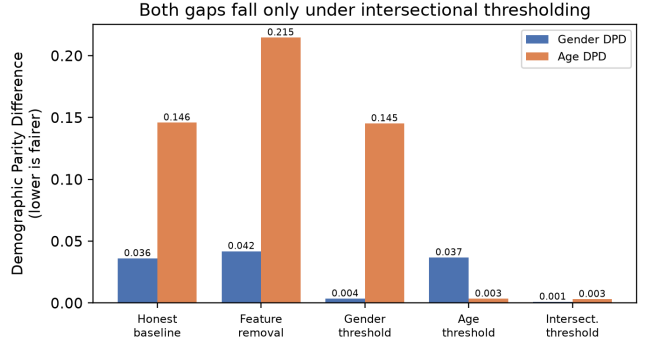


Fig. 2. Demographic parity difference for both protected attributes across the five configurations (seed 42). Feature removal worsens the age gap; single-attribute thresholds close only their own attribute; intersectional thresholding closes both.

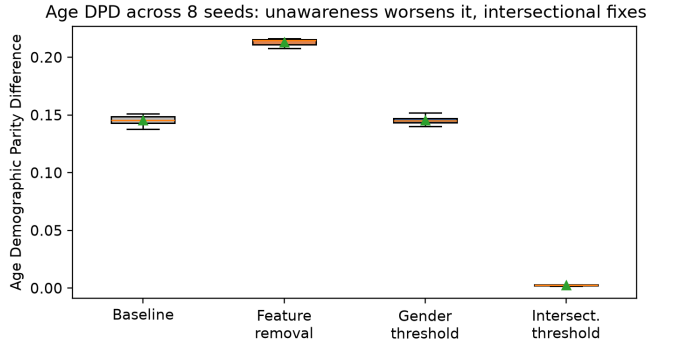


Fig. 3. Age DPD across eight seeds. Fairness through unawareness shifts the gap up, single-attribute gender thresholding leaves it at baseline, and intersectional calibration collapses it to near zero.

cannot speak to non-binary users or to production traffic, and results should not be generalized without re-auditing.

Construct validity. Demographic parity, equal opportunity, and disparate impact can conflict [9]; we report all three and do not claim the mitigated model is “fair” in an absolute sense—only that specific measured gaps change as reported.

VIII. CONCLUSION AND FUTURE WORK

We audited a scalable, interpretable rating classifier on MovieLens 1M for demographic fairness. After correcting a target-leakage flaw, we found a mild gender disparity but a substantial age disparity hidden beneath a 0.72 aggregate accuracy; removing protected attributes significantly worsened the age gap, and single-attribute thresholding closed only the attribute it targeted. We introduced intersectional threshold calibration, which closes the gender *and* age gaps together at a 0.21-point accuracy cost, and validated every headline result across eight seeds with paired significance tests. Future work includes stronger baselines (gradient-boosted trees), regularized intersectional thresholds for many fine-grained strata, in-processing fairness constraints, and extending the audit from rating classification to top- k recommendation.

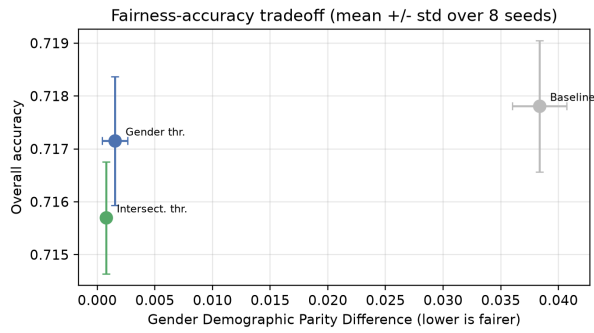


Fig. 4. Gender fairness-accuracy tradeoff (mean $\pm$ std over eight seeds). Threshold calibration reaches near-zero DPD at a small accuracy cost.

REPRODUCIBILITY

All numbers are produced by a single Spark notebook (`Vortex_D4_Fairness.ipynb`) from the raw MovieLens 1M files, over seeds $\{42, 1, 7, 13, 21, 99, 123, 2024\}$, with Spark 3.5.3, Java 17, and Python 3.11. The notebook writes the full numeric record (`fairness_results_d4.json`, `fairness_cross_seed.csv`) and all figures.

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